Lecture 12 // Oct 1 // The Chern-Simons Functional & Reducibles

Last time: We defined the Chern-Simons functional

$$\mathrm{CS}(A) = \frac{1}{8\pi^2}\int_Y \mathrm{tr}\left(A\wedge dA + \frac{2}{3}A\wedge A\wedge A\right)$$

and saw that $\mathrm{grad}_A \mathrm{CS} = *F_A$. We also interpreted $\mathrm{Crit}(\mathrm{CS})$ in two ways:

$$\mathrm{Crit}(\mathrm{CS}) = \{\text{flat connections modulo gauge}\} \leftrightarrow R.$$

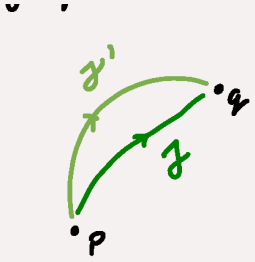
Today, we'll complete the proof of the theorem from last time.

Theorem 1
Critical points of the Chern-Simons functional modulo gauge are in bijection with $\mathrm{SU}(2)$ representations of $\pi_1(Y)$ modulo conjugation.

Proof (continued). We must show injectivity. Let ∇_0, ∇_1 be two connections such that $\mathrm{hol}_p^{\nabla_0}$ and $\mathrm{hol}_p^{\nabla_1}$ are related by conjugation via some fixed $g_p \in \mathrm{SU}(2)$.

We want to show ∇_0 and ∇_1 are gauge equivalent. We construct a global gauge transformation from g_p as follows: for any q, choose a path γ from p to q. Then define

$$g_q = \mathrm{hol}^{\nabla_1}(\gamma)\circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}).$$



Is g_q well-defined? Yes: choose another γ' from p to q. Then $g_q' = \mathrm{hol}^{\nabla_1}(\gamma')\circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')$, and

$$\begin{aligned} g_q g_q'^{-1} &= \left(\mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\right)\circ\left(\mathrm{hol}^{\nabla_1}(\gamma')\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')\right)^{-1} \\ &= \left(\mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\right)\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')^{-1}\circ g_p^{-1}\circ \mathrm{hol}^{\nabla_1}(\gamma')^{-1} \\ &= \mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\circ \mathrm{hol}^{\nabla_0}(\gamma')\circ g_p^{-1}\circ \mathrm{hol}^{\nabla_1}(\bar{\gamma}'). \end{aligned}$$

Since $\mathrm{hol}^{\nabla_0}(\bar{\gamma})\circ \mathrm{hol}^{\nabla_0}(\gamma') \in \mathrm{hol}_p^{\nabla_0}$, and $\mathrm{hol}_p^{\nabla_0}$ & $\mathrm{hol}_p^{\nabla_1}$ are related by $g_p \in \mathrm{SU}(2)$,

$$g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}))\circ \mathrm{hol}^{\nabla_0}(\gamma')\circ g_p^{-1} = \mathrm{hol}^{\nabla_1}(\bar{\gamma})\circ \mathrm{hol}^{\nabla_1}(\gamma'),$$

so

$$g_q g_q'^{-1} = \mathrm{hol}^{\nabla_1}(\gamma) \circ \mathrm{hol}^{\nabla_1}(\bar{\gamma}) \circ \mathrm{hol}^{\nabla_1}(\gamma') \circ \mathrm{hol}^{\nabla_1}(\bar{\gamma}') = \mathrm{id}.$$

By construction, $\mathrm{hol}^{\nabla_0} = g_q^{-1} \circ \mathrm{hol}^0(\gamma) \cdot g_p$, so certainly the parallel transports of ∇_0, ∇_1 behave as if related by a gauge transformation. □

The Instanton Chain Complex

Recall that the generators of the Instanton Chain Complex (ICC) are the critical points of the Chern-Simons functional, and the differential is defined by looking at the gradient flow of CS, $\mathrm{grad}_A(\mathrm{CS}) = *F_A$. But how can we do this on a three-manifold? The general slogan is that studying the gradient flow of CS on a three-manifold Y^3 is the same as studying SD/ASD Yang-Mills connections on $Y^3 \times \mathbb{R}$. There's a very precise connection between the three and four-manifold stories; this slogan is the first piece. Specifically, instead of studying SD/ASD on closed X^4, we looked at the open 4-manifold $\mathbb{R} \times Y$. On $\mathbb{R} \times Y$, consider the trivial SU(2)-bundle $\mathbb{R} \times Y \times \mathbb{C}^2$. Since $c_2(\mathbb{R} \times Y) = 0$. By the Künneth formula, $c_2(\mathbb{R} \times Y) = \sum_{i+j=2} c_i(\mathbb{R}) \otimes c_j(Y) = 0$, this is only one.

Now let's fix a metric g_Y on Y, taking the product metric $dt^2 + g_Y$ on $\mathbb{R} \times Y$. We may use this to orient $\mathbb{R} \times Y$ by declaring a positively-oriented basis to be $\{dt, \underbrace{dx, dy, dz}_{Y \text{ directions}}\}$, with dt in the $\mathbb{R}$ direction.

Let's think about the bundle as a 1-parameter family of objects over Y. A gauge transformation g over $\mathbb{R} \times Y$ is a 1-parameter family of gauge transformations g_t: e.g., a gauge transformation on Y smoothly varying with t.

A connection (rather, a connection 1-form) on $\mathbb{R} \times Y$ is, in general, of the form $A = A_t + B_t\, dt$. This is achieved by separating the dt from dx, dy, dz in the 1-form, so A_t consists of one-forms in Y and B_t is a matrix of 0-forms. Note that A_t, B_t still vary in t, since the x, y, z coordinates are themselves functions of t: $x = x(t), y = y(t), z = z(t)$. Also note that B_t is not a connection on Y.

For each t, define the inclusion map $\iota_t : Y \to \{t\} \times Y \subset \mathbb{R} \times Y$; then A_t is $\iota_t^*(A)$. It will be convenient for us to define the following.

Definition 2
A is in temporal gauge if $B_t = 0$.

Lemma 3
Up to a gauge transformation on $\mathbb{R} \times Y$, every A can be put into temporal gauge.

Proof. We want to find a 1-parameter family of gauge transformations $\{g_t\}$ which kills the dt component. Applying the transformation law, g_t sends B_t to $g_t^{-1} B_t g_t^{-1} - g_t^{-1} dg_t$. Therefore g_t acts on A via

$$A \xmapsto{\sim g_t} g_t^{-1} A g_t + g_t^{-1} dg_t.$$

Simplifying,

$$g_t^{-1}Ag_t + g_t^{-1}dg_t = g_t^{-1}(A_t + B_t\,dt)g_t + g_t^{-1}dg_t = g_t^{-1}A_tg_t + g_t^{-1}B_tg_t\,dt + g_t^{-1}dg_t.$$

The dt component of $g_t^{-1}dg_t$ is $g_t^{-1}\dfrac{\partial}{\partial t}g_t\,dt$, so the dt component of $g_t^{-1}Ag_t + g_t^{-1}dg_t$ is

$$g_t^{-1}B_tg_t\,dt + g_t^{-1}\frac{\partial}{\partial t}g_t\,dt.$$

Therefore we would like to show that for every t, there is a g_t such that

$$g_t^{-1}B_tg_t\,dt + g_t^{-1}\frac{\partial}{\partial t}g_t\,dt = 0.$$

In other words, we want to solve the 1-parameter family of ODEs:

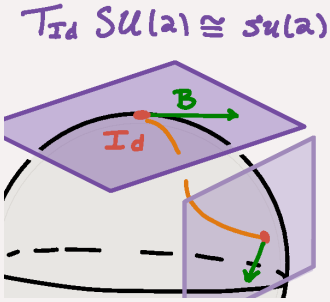
$$g_t^{-1}B_tg_t\,dt + g_t^{-1}\frac{\partial}{\partial t}g_t\,dt = 0 \iff B_tg_t\,dt + \frac{\partial}{\partial t}g_t\,dt = 0 \iff B_tg_t + \frac{\partial}{\partial t}g_t = 0 \iff \frac{\partial}{\partial t}g_t = -B_tg_t.$$

For each fixed $y \in Y$, we want that $\dfrac{\partial}{\partial t}g_t(y) = -B_t(y)g_t(y)$.

Let's interpret the right hand side. Note that $g_t(y)$ is a path in $\mathrm{SU}(2)$, and $B_t(y)$ is a time-varying element of the Lie algebra of $\mathrm{SU}(2)$, $\mathfrak{su}(2)$, which is the tangent space at the identity $T_{\mathrm{Id}}\,\mathrm{SU}(2)$. Explicitly, for a fixed t and y, $B_t(y)$ is a 2×2 matrix in $\mathfrak{su}(2)$, and $g_t(y)$ is a 2×2 matrix in $\mathrm{SU}(2)$.

We can get a new 2×2 matrix in $\mathrm{SU}(2)$ by right-multiplying $B_t(y)$ by $g_t(y)$:

$$B_t(y)\cdot g_t(y) = \begin{pmatrix} b_{11}(t,y) & b_{21}(t,y) \\ b_{12}(t,y) & b_{22}(t,y) \end{pmatrix} \cdot \begin{pmatrix} g_{11}(t,y) & g_{21}(t,y) \\ g_{12}(t,y) & g_{22}(t,y) \end{pmatrix}.$$



Doing so at each t, still fixing y, gives us the pushforward of $B_t(y)$ under right multiplication by $g_t(y)$, defining a vector field over all of $\mathrm{SU}(2)$.

Combining this with the left-hand side, if this vector field equals $\dfrac{d}{dt}g_t(y)$, the path must be an integral curve (e.g., a flowline) for this vector field. By ODE theory, this always exists; if the initial conditions are fixed, it is unique. □

Up to a gauge transformation, we can think of ∇ on $\mathbb{R}\times Y$ as $\{\nabla_t\}$ on Y.

We're now able to ask: what do the SDYM equations look like? First, the self-duality equation is $F_A = *F_A$.

Let $A = A_t$ be in temporal gauge. Calculating $F_A = dA + A\wedge A$, note that it's not true that the 4-dimensional curvature is a 1-parameter family of 3-dimensional curvatures. Even though the connection

itself has no dt component, the entries of this matrix are functions in all four variables, i.e., these entries are allowed to vary in t. This means that dA will have a $\frac{\partial}{\partial t}A_t\,dt$ component, which encodes how the curvature of Y changes in the time direction.

$$dA = dt \wedge \frac{\partial}{\partial t}A_t + \left(\sum_{i=1}^{3} dx_i \wedge \frac{\partial}{\partial x_i}A_t\right) = -\frac{\partial}{\partial t}A_t \wedge dt + \underbrace{\left(\sum_{i=1}^{3} dx_i \wedge \frac{\partial}{\partial x_i}A_t\right)}_{d^{(3)}A_t}$$

$$\Rightarrow F_A = -\frac{\partial}{\partial t}A_t \wedge dt + d^{(3)}A_t + A_t \wedge A_t.$$

For notational convenience, define $F_{A_t} := d^{(3)}A_t + A_t \wedge A_t$. Then $F_A = -\frac{\partial}{\partial t}A_t \wedge dt + F_{A_t}$.

To calculate $*F_A$, let's find the relationship between the three and four-dimensional Hodge stars. There are two main cases: if a 2-form has a time component, and if it doesn't.

1. $*^{(4)}(dx_1 \wedge dt) = -dx_2 \wedge dx_3 = -*^{(3)} dx_1$,
2. $*^{(4)}(dx_1 \wedge dx_2) = -dx_3 \wedge dt = -*^{(3)} (dx_1 \wedge dx_2) \wedge dt$.

This generalizes to the following relationship: let $i, j, k \in \{1, 2, 3\}$.

1. $*^{(4)}(dx_i \wedge dt) = -dx_j \wedge dx_k = -*^{(3)} dx_i$,
2. $*^{(4)}(dx_i \wedge dx_j) = -dx_k \wedge dt = -*^{(3)} (dx_i \wedge dx_j) \wedge dt$.

Thus $*^{(4)}F_A$ can be expressed using $*^{(3)}$ as

$$*^{(4)}F_A = *^{(4)}\left(-\frac{\partial}{\partial t}A_t \wedge dt + F_{A_t}\right) = *^{(4)}\left(-\frac{\partial}{\partial t}A_t \wedge dt\right) + *^{(4)}F_{A_t} = *^{(3)}\frac{\partial}{\partial t}A_t - *^{(3)}F_{A_t} \wedge dt.$$

In full, the SDYM equation becomes:

$$F_A = *F_A.$$
$$-\frac{\partial}{\partial t}A_t \wedge dt + F_{A_t} = *^{(3)}\frac{\partial}{\partial t}A_t - *^{(3)}F_{A_t} \wedge dt$$
$$\frac{\partial}{\partial t}A_t = *^{(3)}F_{A_t}.$$

For $A = A_t$ in temporal gauge, A is a SD YM connection iff A_t follows $\operatorname{grad}_B \mathrm{CS} = *F_B$. Similarly, A is ASD iff A_t is a flow of $-\operatorname{grad}_B \mathrm{CS}$.

Now that we've made sense of what it means to be a generator of the ICC, there's one technical point we must discuss. Just like in the proof of Donaldson's theorem, the space $\mathcal{A}/\mathcal{G}$ in the 3-dimensional setting is not a manifold, since there can be connections with nontrivial stabilizer. Recall that we proved that the stabilizer of a connection is the centralizer of the connection. This holds in the 3-d case too.

Subgroups of SU(2)	Centralizer
I	SU(2)
$-I$	SU(2)
S^1 or a conjugate of S^1	S^1
Subgroups of S^1 or conjugate	S^1
non-abelian	$\pm I$
SU(2)	$\pm I$

Lemma 4
Flat connections with stabilizer larger than $\pm I$ correspond bijectively to abelian representations of $\pi_1(Y)$, i.e., those which factor through $H_1(Y;\mathbb{Z})$.

Definition 5
∇ is called reducible if $\mathrm{Stab}(\nabla)$ is larger than $\pm I$.

This is the first place we'll use that Y^3 is a $\mathbb{Z}\mathrm{HS}^3$. If Y is a homology sphere, there is only one reducible flat connection, which corresponds to the trivial representation into SU(2).

Remark 6
If Y is a $\mathbb{Q}\mathrm{HS}^3$, could have torsion. Computability of I breaks down based on how computable the character variety is, but also on how computable the differential is. The singular points aren't necessarily isolated, but they can be post perturbation.